

Coherent, Not Merely Accurate

Scoring the no-arbitrage constraint reveals cross-strike structure that RMSE misses by an order of magnitude

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Abstract

Prediction-market venues list *ladders*: families of contracts on one underlying event at different thresholds. A ladder carries something a single price series cannot — a mechanical no-arbitrage constraint. Weather temperature bins form a *partition*, so probabilities must sum to one; sports over/under totals form a *CDF*, so probability must be non-increasing in the threshold. The constraint is a property of the forecast alone, so it can be scored without labels, and no baseline satisfies it by construction.

Scoring it changes what the evaluation says. A cross-strike model significantly outperforms an otherwise-identical single-strike control on RMSE, on every seed under a bootstrap resampling ladders — but the effect is small (−1.28% weather, −2.85% sports). Measured on the constraint, the same models separate by −2.92% and −28.56%: 2.3× and 10.0× the accuracy effect. The cross-strike model removes 36% of the violations persistence inherits, against the control’s 10%. The axis also discriminates where accuracy cannot: where the per-seed RMSE gate fails on two of five seeds, the coherence gate passes on all five. **Accuracy-only evaluation understates cross-strike modelling by close to an order of magnitude.**

Two ablations establish the gain is *information*, not capacity: shuffling strike order costs only 8.3% of the edge, while replacing neighbours with unrelated ones makes the model worse than one that never had neighbours. We report the constraint’s economics honestly — the two metrics are different kinds of quantity, and neither is tradeable — and measure a ladder overround of 2.85% compressing to 1.20% by the last observed price. Finally, a label-free joint-embedding representation reaches parity with a supervised encoder under an identical frozen probe, with two by-products cutting against standard practice: effective rank is *anti*-correlated with probe quality here, and selecting checkpoints on the self-supervised loss costs ~8 points.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Applied computing** → *Economics*.

Keywords

prediction markets, no-arbitrage constraints, evaluation, cross-sectional forecasting, self-supervised learning

1 Introduction

Forecasting quality in finance is almost always reported as accuracy: RMSE, MAE, a directional hit rate. On a single price series that is close to the only option. But a *ladder* — a family of contracts on one

underlying event at different thresholds — admits a second, independent axis, because the joint forecast must satisfy an arithmetic constraint that has nothing to do with what actually happened.

Two geometries occur naturally on prediction-market venues. Weather temperature ladders are a **partition**: single-degree bins plus two tail thresholds, mutually exclusive and exhaustive, so the probabilities form a density summing to approximately one. Sports over/under totals are a **CDF**: nested thresholds on one game, so $P(\text{total} > x)$ must be non-increasing in x . Violating either is a riskless inconsistency in the forecast itself.

This has three properties that make it valuable as an evaluation axis. It is *label-free* — coherence is a function of the prediction alone, so it can be computed before any outcome is known. It is *unsatisfiable by construction* for a single-strike model, which never sees the other strikes and so cannot know what it is violating. And it is *economically interpretable*: a monotonicity violation on a CDF ladder is a lock, quotable in dollars per share.

Our central finding is that this axis is not merely a second opinion — it is a substantially sharper instrument:

On identical models, identical data and an identical bootstrap, the cross-strike effect measured on the no-arbitrage constraint is 2.3× the effect measured on RMSE for partition ladders and 10.0× for CDF ladders. In a configuration where the pre-registered per-seed RMSE gate fails, the coherence gate still passes on every seed.

Contributions. We define a coherence evaluation axis for both ladder geometries and release its implementation (Section 4). We show a cross-strike model beats a matched single-strike control on both axes and both ladder families, with parameter budgets within 0.15% (Sections 5–6). We establish by two ablations that the gain is cross-strike *information*, not capacity (Section 7). We report the constraint’s economics without overclaiming, including a units trap that is easy to fall into (Section 8). And we show a label-free representation reaches parity with a supervised one on this task, with two negative by-products of wider interest (Section 9).

2 Related work

Coherence and no-arbitrage in forecasting. Requiring probabilistic forecasts to respect known structure is long-established: hierarchical forecast *reconciliation* projects incoherent base forecasts onto an aggregation constraint [4, 8], and isotonic regression enforces monotonicity [2]. That literature treats coherence as something to *impose* at post-processing. We use it differently — as a *measurement* that discriminates between models, applied to unprojected

forecasts. The distinction matters: reconciliation would make every model trivially coherent and destroy the signal we report.

Prediction markets. The economic literature establishes that market prices aggregate dispersed information and forecast outcomes competitively [3, 9], and studies internal consistency and arbitrage relations across related contracts [6, 10]. Our contribution is not that ladders should be coherent — that is arithmetic — but that the *degree* of incoherence is a usable model-selection signal with far more resolution than accuracy.

Cross-sectional and self-supervised sequence models. Multivariate forecasting exploits cross-series structure [7, 11], and joint-embedding predictive architectures learn representations by predicting in latent space rather than reconstructing [1, 5]. Section 9 tests the latter in the setting where its premise most clearly applies — per-strike microstructure noise is exactly the nuisance detail, while the shared latent ladder state survives.

3 Ladders and their two geometries

A ladder is a set of contracts $\{c_1, \dots, c_K\}$ on one underlying event, indexed by an ordered threshold. We build panels from resolved Polymarket markets: the task is unchanged from the univariate setting — given $C = 30$ observed probabilities forecast the next $H = 10$ — except that a model may condition on all strikes in the ladder simultaneously.

The two families are not superficial variants of each other. Measured at the forecast origin, before any model is trained:

Table 1: The two ladder geometries are genuinely different objects.

Family	Geometry	Fully monotone at origin
Weather bins	partition (density)	7.9%
Sports totals	CDF (nested)	86.4%

A partition is almost never monotone, and should not be — a temperature density peaks in the middle. A CDF almost always is. So the two families test different constraints, and a result that appears on both is a statement about cross-strike structure rather than about one constraint’s idiosyncrasies.

4 The coherence axis

For each family we score the constraint the geometry implies. Both are computed per window and per horizon step, on raw model output with no projection.

Partition ladders — mass normalisation. For a predicted ladder $\hat{p} \in \mathbb{R}^K$ with realised y , the error is the deviation of total predicted mass from total realised mass:

$$E_{\text{mass}} = \left| \sum_k \hat{p}_k - \sum_k y_k \right|.$$

CDF ladders — monotonicity. With strikes ordered by increasing threshold, $P(\text{total} > x)$ must be non-increasing, so any positive step is a violation. We report both the rate and the *magnitude*, since only the magnitude is quotable in dollars:

$$V = \sum_k \max(\hat{p}_{k+1} - \hat{p}_k, 0).$$

Why persistence is the right zero. Persistence predicts no drift, so its mass error is exactly the *realised* drift — a quantity fixed by the data, independent of any model. This makes the axis auditable: we verified our evaluator reproduces the independently measured persistence drift (0.04009868 against 0.04009867) before scoring any checkpoint. An evaluation axis that cannot be reconciled to a pre-model measurement should not be trusted.

Uncertainty. All intervals come from a cluster bootstrap that re-samples **ladders**, not windows. Windows within a ladder share an underlying event and are strongly dependent; resampling them independently would understate the interval. Significance is reported **per seed** — every individual seed’s CI must exclude zero — and we never claim an effect on a pooled or ensembled interval.

5 Models

The comparison is between two models that differ in exactly one respect: whether cross-strike information is reachable.

- **SingleStrikeControl** — forecasts each strike from its own history only.
- **CrossStrikePanel** — identical, but attends across strikes within a ladder before forecasting.

Fair-comparison discipline. Identical input and output tensors, identical total layer count, parameter budgets within **0.15%**, and an identical residual head with persistence nested as a special case. This matters because a previous null in this project on a different architecture comparison was uninterpretable precisely because the two arms were not matched on capacity; a capacity-confounded “win” and a capacity-confounded “null” are equally worthless.

6 Results

6.1 Cross-strike information improves accuracy

Table 2: RMSE, both ladder families. Panel and control differ only in whether cross-strike information is reachable. Bootstrap resamples ladders. Both families are the re-run checkpoints, which are the models the coherence axis is scored on; the original sports run is reported separately below.

	Weather (3 seeds)	Sports (5 seeds)
Persistence	0.038748	0.076427
SingleStrikeControl	0.038457	0.075731
CrossStrikePanel	0.037963	0.073573
Panel vs control (rel.)	−1.28%	−2.85%

On weather, the per-seed differences panel–control are -0.000533 , -0.000471 and -0.000478 , every seed’s CI excluding zero. Seed standard deviations are 0.000029 (control) and 0.000029 (panel).

Sports: report both runs. The sports gate is instructive and we do not tidy it. In the original run (control 0.075701 ± 0.000066 , panel 0.074046 ± 0.000898), four of five seeds were strongly significant and clustered within 0.0003 at an effect roughly 4× weather’s, while seed 7 landed at $+0.000038$ (CI $[-0.00028, +0.00034]$) — exactly

control level. That **fails** the pre-registered per-seed gate, and we reported it as a failure rather than dropping the seed.

Re-running with identical data, width and seeds in a different environment, all five seeds converge (-0.00207 to -0.00227 , every CI excluding zero) and the gate **passes**. The panel’s across-seed standard deviation collapses from 0.000898 to 0.000067 , a $13\times$ tightening — which is what a single non-converged seed looks like once it converges. The honest reading is that a per-seed gate on 1,322 training ladders sits near the edge of its own stability, and that a single-seed failure which vanishes on re-run is a statement about optimisation variance rather than about the data. We report both runs and do not silently replace the first.

Table 3: Sports, original run: per-seed panel–control dRMSE with ladder-level bootstrap CIs. Seed 7 fails the gate and is reported, not dropped.

Panel seed	dRMSE vs control	95% CI	Gate
17	-0.002099	$[-0.00277, -0.00144]$	pass
29	-0.002062	$[-0.00281, -0.00130]$	pass
43	-0.001914	$[-0.00261, -0.00121]$	pass
59	-0.002235	$[-0.00296, -0.00153]$	pass
7	$+0.000038$	$[-0.00028, +0.00034]$	fail

A hypothesis we refuted. The natural explanation for the seed-7 failure was data volume. We tested it by halving the ladder width, which nearly doubles the number of ladders. It made the RMSE gate *worse*, not better (Table 6): with only one neighbour the effect itself weakens from ~ -0.00215 to ~ -0.00096 , and the weaker effect dominates the larger sample. Data volume was not the binding constraint.

6.2 The coherence axis is a sharper instrument

Scoring the same forecasts on the constraint instead:

Table 4: Coherence. Weather is mass-normalisation error; sports is monotonicity violation magnitude. Lower is better; every seed’s CI excludes zero.

Family	Metric	Pers.	Control	Panel	Gate
Weather	mass error	0.040099	0.038272	0.037153	3/3
Sports	violation	0.010120	0.009077	0.006485	5/5

The cross-strike model removes **36%** of the no-arbitrage violations that persistence inherits; the control removes only **10%**. Put beside the accuracy effect:

Table 5: The same model comparison, measured two ways.

Family	RMSE effect	Coherence effect	Ratio
Weather	-1.28%	-2.92%	$2.3\times$
Sports	-2.85%	-28.56%	$10.0\times$

The discriminating case. A ratio could be dismissed as a scaling artifact of two differently normalised metrics. The stronger evidence is that the two axes *disagree about the verdict*. At halved ladder width:

Table 6: Same models, same data, same bootstrap — opposite conclusions. Width 2 is where the axes separate.

Width	Train ladders	RMSE gate	Coherence gate
3 (original run)	1,322	4/5 — fail	not measured
3 (re-run)	1,322	5/5 — pass	5/5 — pass
2	2,522	3/5 — fail	5/5 — pass

At width 2 an accuracy-only evaluation concludes the cross-strike effect is not established, while the coherence evaluation concludes it is, on all five seeds. The coherence gate passes in *every* configuration we ran, including both where RMSE fails. That is the practical claim: the axis is not a restatement of accuracy, and it resolves effects that accuracy cannot.

6.3 The effect was predicted before any model was trained

Ahead of training we measured that ladder total-mass drift correlates -0.31 with observable history. Total mass is an order-invariant aggregate that a single-strike model structurally cannot observe, so this predicted both that a cross-strike model should gain and *what kind* of information it should exploit. The ablations below confirm it does exactly that.

7 Information, not capacity

The obvious objection is that the panel simply has more effective capacity. Two ablations rule that out.

Strike-order shuffle. Randomly permuting strike order within a ladder destroys only **8.3%** of the edge. The panel does not rely on adjacency; it uses an order-invariant aggregate, exactly as the -0.31 drift correlation predicted.

Donor neighbours. We replace the non-centre strikes with strikes drawn from unrelated windows, so every marginal remains in-distribution and only the *relationship* is destroyed. Centre-strike RMSE moves from 0.045720 (-1.78% vs control) to 0.048378 — $+3.93\%$ vs control.

With fake neighbours the panel is worse than a model that never had neighbours at all. Extra capacity cannot produce that: capacity does not become a liability when its input is uninformative, but a model that has learned to rely on a genuine cross-strike relation is actively misled when the relation is broken.

8 What the constraint is worth, and what it is not

A units trap, which we fell into first. The two coherence metrics are different *kinds* of quantity and must not be scaled against a common threshold. Mass-normalisation error is a **forecast error** and is not money; its natural comparator is the univariate RMSE edge (0.0016 in probability units), against which 0.040 of drift means

roughly 25× more error is available cross-sectionally than along a single series. A monotonicity violation is P&L — it is a lock — so its comparator is execution cost.

An earlier draft of this work scaled both against a single-leg cost threshold and reported them as “20–29× tradeable”. That was wrong on both halves. We state it because the error is easy to make and the corrected reading is much less exciting.

Corrected reading. The mean violation magnitude is 0.010120 for persistence, i.e. **1.01 modal ticks** on a market whose modal price increment is 0.0100. Capturing it requires a two-leg trade that must clear roughly two spread crossings. It is **borderline at best, and this is not a trading result**. The accuracy gain is smaller still: 0.00055 in probability units, below the ~ 0.0014 round-trip cost a companion univariate study measures as the ceiling on this venue’s directional edge. The contribution here is methodological.

An observed microstructure fact. Full-ladder mass at the forecast origin is 1.0285 — a **2.85% overround** — compressing to 1.0120 (**1.20%**) by the last observed price. The margin narrows as the event resolves. Overround is the market’s margin rather than free money, but the *compression* is a measured regularity we have not seen reported elsewhere, and it is what makes total-mass drift partially predictable.

Scale caution. Panel-scale RMSE (~ 0.0387) is not comparable to single-series prediction-market RMSE (~ 0.10), because temperature bins carry low probability. The two must not be quoted side by side unnormalised.

9 A label-free representation reaches parity

Ladders are the setting where the joint-embedding premise most clearly applies: a single scalar probability path has no high-dimensional nuisance detail to discard, whereas a ladder’s per-strike microstructure noise is exactly the nuisance while the shared latent state survives.

Under the canonical protocol — latent pretraining, frozen encoder, linear probe — on the test split (13,189 windows / 1,390 ladders):

Table 7: Frozen-representation comparison, test MSE.

	Test MSE
Persistence	0.00150140
Label-free JEPA + linear probe	0.00144400
Supervised encoder, same probe	0.00145084
Supervised model, trained head (ceiling)	0.00143702

The label-free representation reaches **89.2%** of the fully supervised model’s gain over persistence. Against the supervised *encoder* under the ladder-level bootstrap, 1 of 3 seeds is significant and 2 are not, so the claim is **parity, not superiority**. It also confers no advantage when labels are available: initialising from it and fine-tuning is indistinguishable from training from scratch (mean difference 1.4×10^{-7} , all seeds n.s.).

Table 8: How the label-free representation got there (validation share of the supervised gain). Both decisive changes came from the literature; every intervention we designed from our own collapse diagnosis made things worse.

Version	Change	Share
v2	variance/covariance reg. on encoder output	50.8%
v3	isotropy reg. on encoder output	53.3%
v4	masked-strike objective + cosine LR	61.9%
v5	regularisation moved to a projector	71.1%
v6	shorter cosine schedule	75.3%

We are at the benchmark’s resolution limit. The gap from the label-free model to the supervised head is 6.98×10^{-6} , while bootstrap CIs are $\pm 2 \times 10^{-5}$ — about 3× wider. This test set cannot resolve differences at that margin, so chasing a win via frozen probes would mean optimising an unmeasurable quantity. We state the limit rather than reporting the point estimate as a result.

Two by-products that cut against standard practice. First, **effective rank is anti-correlated with probe quality here**. Regularising the encoder directly gives effective rank 98/128 and 53.3% of the supervised gain; moving the same regularisation to a projector leaves the encoder at rank 1.4/128 and gives 71%+. An 18-point gap in favour of the collapsed-*looking* representation. This inverts the usual collapse heuristic, and it has a clean explanation: the ladder signal genuinely lives in about two dimensions (level and mass), so forcing isotropy on the representation dilutes it. Second, **selecting checkpoints on the self-supervised loss costs ~ 8 points**, confirmed on two independent runs — in this setting the SSL loss is non-monotone against probe performance.

Methodology note. This claim was wrong three times before it was right, at 54.9%, 2.5% and 50.8%, and each earlier figure was a measurement artifact rather than a property of the method: a probe seeded from an encoder that had already seen forecast supervision; an MLP probe head that failed to optimise and converged to persistence; and validation-selected figures reported without a test-set bootstrap. Each was caught by a control that took minutes and had been skipped. We adopted the rule that frozen-representation claims require a linear probe, every seed, the same probe applied to a supervised encoder as a control, and a test-set bootstrap before any comparative claim.

10 Limitations

Two ladder families, one venue. Weather and sports are structurally different, which is the point, but both come from one exchange. The coherence axis is defined by geometry rather than by venue and should transfer, but we have not shown that.

The sports gate is near its stability edge. A per-seed gate on 1,322 training ladders produced a failure that did not reproduce. We report both runs; a reader should treat the sports RMSE effect as established with less margin than the weather one. The *coherence* gate passed in every configuration, including both where RMSE failed.

Coherence is measured, not imposed. We deliberately do not project forecasts onto the constraint. Projection would make every model coherent, remove the signal, and is a different contribution. Whether reconciliation improves accuracy here is untested.

Not a trading result. See Section 8. The violation magnitude is roughly one modal tick against a two-leg capture, and the accuracy gain is well below round-trip cost.

The panel comparison is capacity-matched, not architecture-exhaustive. Parameter budgets agree within 0.15% and layer counts are identical, but we test one cross-strike mechanism, not the space of them.

11 Conclusion

Ladders give prediction-market forecasting something a single series cannot: an arithmetic constraint the forecast must satisfy, independent of any outcome. We show that scoring it is not a formality. On the same models, the same data and the same bootstrap, the cross-strike effect is $2.3\times$ larger on partition ladders and $10.0\times$ larger on CDF ladders when measured as coherence rather than accuracy — and in the configuration where the two axes disagree, coherence resolves an effect that RMSE cannot.

The mechanism is cross-strike information rather than capacity: shuffling strike order costs almost nothing, while replacing neighbours with unrelated ones makes the model worse than never having had them. None of it is tradeable, and we measure by how much rather than asserting it.

The general point is that when a forecasting problem carries structure, the structure is an evaluation axis and not only a modelling opportunity. It is label-free, it is unsatisfiable by construction for models that cannot see the structure, and here it has close to an order of magnitude more resolution than the accuracy metric it sits beside.

Acknowledgments

To be added after review.

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